FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- MCQ

CLASS XII

PHYSICS

ATOMS

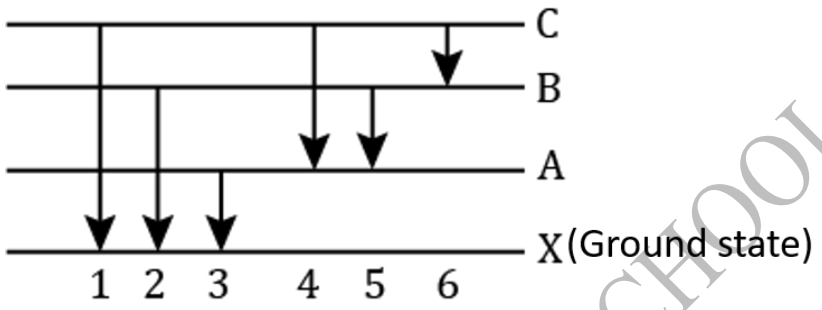
---

1. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Identify the correct statement.
   (a) its kinetic energy increases and its potential and total energies decrease
   (b) its kinetic energy decreases, potential energy increases and its total energy remains the same
   (c) its kinetic and total energies decrease and its potential energy increases
   (d) its kinetic, potential and total energies decrease
2. An alpha particle of energy ½mv^2 bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to
   (a) $\frac{1}{Ze}$ (b) v^2 (c) $\frac{1}{m}$ (d) $\frac{1}{v^4}$
3. The Bohr model of an atom
   (a) assumes that the angular momentum of electrons is quantized
   (b) uses Einstein's photoelectric equation
   (c) predicts continuous emission spectra for atoms
   (d) predicts the same emission spectra for all types of atoms
4. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the third orbit, the corresponding wavelength of the photon will be
   (a) $\frac{20}{7}$ λ (b) $\frac{20}{13}$ λ (c) $\frac{16}{25}$ λ (d) $\frac{9}{16}$ λ
5. An electron in the hydrogen atom jumps from excited state n to the ground state. The wavelength so emitted illuminates a photosensitive material having work function 2.75 eV. If the stopping potential of the photoelectron is 10 V, then the value of n is
   (a) 5 (b) 2 (c) 3 (d) 4
6. Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is
   (a) $\frac{5}{27}$ (b) $\frac{3}{23}$ (c) $\frac{7}{29}$ (d) $\frac{9}{31}$
7. The ionization energy of the electron in the hydrogen atom in its ground state is 13.6 eV. The atoms are excited to higher energy levels to emit radiations of 6 wavelengths. Maximum wavelength of emitted radiation corresponds to the transition between
   (a) n= 3 and n= 1 states (b) n= 2 and n= 1 states
   (c) n= 4 and n= 3 states (d) n= 3 and n=2 states
8. Out of the following which is the energy by which it is not possible for a photon to be emitted by a hydrogen atom according to Bohr's atomic model?
   (a) 13.6 eV (b) 0.65 eV (c) 1.9 eV (d) 11.1 eV
9. When an electron makes transition from n= 4 to n=2, then emitted line spectrum will be

FREEDOM INTERNATIONAL SCHOOL

(a) first line of Lyman series (b) second line of Balmer series
(c) first line of Paschen series (d) second line of Paschen series

10. The figure indicates the energy level diagram of an atom and the origin of six spectral lines in emission (e.g., line no. 5 arises from the transition from level B to A). Which of the following spectral lines will not occur in the absorption spectrum?



(a) 4, 5, 6 (b) 1, 2, 3, 4, 5, 6 (c) 1, 2, 3 (d) 1, 4, 6

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** Bohr had to postulate that the electrons in stationary orbits around the nucleus do not radiate.
Reason: According to classical physics all moving electrons radiate.
12. **Assertion:** Hydrogen atom consists of only one electron but its emission spectrum has many lines.
Reason: Only the Lyman series is found in the absorption spectrum of a hydrogen atom whereas in the emission spectrum, all the series are found.
13. **Assertion:** Atoms of each element are stable and emit characteristic spectrum.
Reason: The spectrum provides useful information about the atomic structure.
14. **Assertion:** In alpha particle scattering, the number of alpha- particle undergoing head on collision is small.
Reason: Small fraction of the number of incident particles rebound.
15. **Assertion:** Bohr model cannot be extended to two or more electron atoms.
Reason: Each electron in the atom interacts not only with the positively charged nucleus but also with all other electrons.

FREEDOM INTERNATIONAL SCHOOL